LANDSLIDES, AVALANCHES AND GLOF RISK

MOUNTAIN-HAZARD TAXONOMY → SUSCEPTIBILITY-HAZARD-RISK LADDER → SLOPE-FAILURE MECHANISM → CASCADE MAP → INSTITUTIONAL OWNER MATRIX → GLACIAL-LAKE MONITORING RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00

STAGE 00

MOUNTAIN-HAZARD TAXONOMY

MOUNTAIN-HAZARD TAXONOMY · ROCK DEBRIS OR EARTH MOVEMENT · DETACHED ROCK FALLS · DEBRIS FLOW · WATER-RICH CHANNELISED MASS · SNOWPACK RELEASE · LAKE OUTBURST FLOOD

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
LANDSLIDE → rock debris or earth movement	DEBRIS FLOW → water-rich channelised mass	LANDSLIDE → rock debris or earth movement
ROCKFALL → detached rock falls / bounces / rolls	AVALANCHE / GLOF → snowpack release / lake outburst flood	ROCKFALL → detached rock falls / bounces / rolls

HAZARD / ACTOR / INSTRUMENT
- LANDSLIDE → rock debris or earth movement
- ROCKFALL → detached rock falls / bounces / rolls

DECISIVE RISK OR MANDATE DISTINCTION
- DEBRIS FLOW → water-rich channelised mass
- AVALANCHE / GLOF → snowpack release / lake outburst flood

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- LANDSLIDE → rock debris or earth movement
- ROCKFALL → detached rock falls / bounces / rolls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVALANCHE / GLOF → snowpack release / lake outburst flood.

01

STAGE 01

SUSCEPTIBILITY-HAZARD-RISK LADDER

SUSCEPTIBILITY-HAZARD-RISK LADDER · WHERE TERRAIN MAY FAIL · PROBABILITY TIMING INTENSITY · EXPOSURE + VULNERABILITY + CAPACITY · EXPECTED HARM AND DISRUPTION

SUSCEPTIBILITY → WHERE terrain may fail → HAZARD → probability timing intensity → EXPOSURE + VULNERABILITY + CAPACITY → RISK → expected harm and disruption

RISK CONDITION / INPUT
- SUSCEPTIBILITY → WHERE terrain may fail
- HAZARD → probability timing intensity

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- EXPOSURE + VULNERABILITY + CAPACITY
- RISK → expected harm and disruption

LAST-MILE / INCLUSION / OUTCOME LIMIT
- SUSCEPTIBILITY → WHERE terrain may fail
- HAZARD → probability timing intensity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPOSURE + VULNERABILITY + CAPACITY; RISK → expected harm and disruption.

02

STAGE 02

SLOPE-FAILURE MECHANISM

SLOPE-FAILURE MECHANISM · GEOLOGY SLOPE WEATHERING DRAINAGE LAND USE · RAIN QUAKE EROSION CUTTING LOADING · SLIDE FALL OR FLOW · ROUTE SETTLEMENT SERVICE CONSEQUENCE

CONDITIONING → geology slope weathering drainage land use → TRIGGER → rain quake erosion cutting loading → FAILURE → slide fall or flow → EXPOSURE → route settlement service consequence

RISK CONDITION / INPUT
- CONDITIONING → geology slope weathering drainage land use
- TRIGGER → rain quake erosion cutting loading

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- FAILURE → slide fall or flow
- EXPOSURE → route settlement service consequence

LAST-MILE / INCLUSION / OUTCOME LIMIT
- CONDITIONING → geology slope weathering drainage land use
- TRIGGER → rain quake erosion cutting loading

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONDITIONING → geology slope weathering drainage land use.

03

STAGE 03

CASCADE MAP

CASCADE MAP · RIVER BLOCKAGE · BLOCKAGE OR LAKE-DAM FAILURE · OUTBURST FLOOD · CLOUD BURST · FLASH FLOW + DEBRIS + SLOPE FAILURE

EARTHQUAKE / RAIN → LANDSLIDE → LANDSLIDE → RIVER BLOCKAGE → BLOCKAGE OR LAKE-DAM FAILURE → OUTBURST FLOOD → CLOUD BURST → FLASH FLOW + DEBRIS + SLOPE FAILURE

HAZARD / RISK / RESILIENCE CONCEPT
- EARTHQUAKE / RAIN → LANDSLIDE
- LANDSLIDE → RIVER BLOCKAGE

INSTITUTIONAL OR COMMUNITY RESPONSE
- BLOCKAGE OR LAKE-DAM FAILURE → OUTBURST FLOOD
- CLOUD BURST → FLASH FLOW + DEBRIS + SLOPE FAILURE

CAPACITY / STATUS / EVIDENCE LIMIT
- EARTHQUAKE / RAIN → LANDSLIDE
- LANDSLIDE → RIVER BLOCKAGE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOUD BURST → FLASH FLOW + DEBRIS + SLOPE FAILURE.

04

STAGE 04

INSTITUTIONAL OWNER MATRIX

INSTITUTIONAL OWNER MATRIX · LANDSLIDE MAPS INVENTORY RESEARCH FORECAST DEVELOPMENT · SNOW AND AVALANCHE WARNINGS · EARTH OBSERVATION AND INVENTORIES · DOWNSTREAM WATER AND PUBLIC ACTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
GSI → landslide maps inventory research forecast development	NRSC / ISRO → Earth observation and inventories	GSI → landslide maps inventory research forecast development
DGRE / DRDO → snow and avalanche warnings	CWC / STATE / DISTRICT → downstream water and public action	DGRE / DRDO → snow and avalanche warnings

HAZARD / ACTOR / INSTRUMENT
- GSI → landslide maps inventory research forecast development
- DGRE / DRDO → snow and avalanche warnings

DECISIVE RISK OR MANDATE DISTINCTION
- NRSC / ISRO → Earth observation and inventories
- CWC / STATE / DISTRICT → downstream water and public action

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- GSI → landslide maps inventory research forecast development
- DGRE / DRDO → snow and avalanche warnings

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CWC / STATE / DISTRICT → downstream water and public action.

05

STAGE 05

GLACIAL-LAKE MONITORING RAIL

GLACIAL-LAKE MONITORING RAIL · 1 REPEAT SATELLITE SCREENING · 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER · 3 FIELD VALIDATION + DOWNSTREAM EXPOSURE · 4 WARNING EVACUATION AND REVIEW

1 REPEAT SATELLITE SCREENING → 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER → 3 FIELD VALIDATION + DOWNSTREAM EXPOSURE → 4 WARNING EVACUATION AND REVIEW

HAZARD / RISK / RESILIENCE CONCEPT
- 1 REPEAT SATELLITE SCREENING
- 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER

INSTITUTIONAL OR COMMUNITY RESPONSE
- 3 FIELD VALIDATION + DOWNSTREAM EXPOSURE
- 4 WARNING EVACUATION AND REVIEW

CAPACITY / STATUS / EVIDENCE LIMIT
- 1 REPEAT SATELLITE SCREENING
- 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER.

06

STAGE 06

ZONATION-TO-DECISION BRIDGE

ZONATION-TO-DECISION BRIDGE · HIGHER SUSCEPTIBILITY · AVOID OR REQUIRE SITE INVESTIGATION · CONTROL LAND USE DRAINAGE CUTTING LOADING · MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

HIGHER SUSCEPTIBILITY / HAZARD? → AVOID OR REQUIRE SITE INVESTIGATION → CONTROL LAND USE DRAINAGE CUTTING LOADING → MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

HAZARD / RISK / RESILIENCE CONCEPT
- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

INSTITUTIONAL OR COMMUNITY RESPONSE
- CONTROL LAND USE DRAINAGE CUTTING LOADING
- MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

CAPACITY / STATUS / EVIDENCE LIMIT
- HIGHER SUSCEPTIBILITY / HAZARD?
- AVOID OR REQUIRE SITE INVESTIGATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTROL LAND USE DRAINAGE CUTTING LOADING; MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK.

07

STAGE 07

SLOPE-MEASURE PORTFOLIO

SLOPE-MEASURE PORTFOLIO · SUBSURFACE DRAINAGE · CONTROLLED EXCAVATION AND TOE PROTECTION · RETAINING PROTECTION BIOENGINEERING · SITE-SPECIFIC DESIGN MAINTENANCE RESIDUAL RISK

WATER → surface / subsurface drainage → GEOMETRY → controlled excavation and toe protection → MATERIAL → retaining protection bioengineering → LIMIT → site-specific design maintenance residual risk

HAZARD / RISK / RESILIENCE CONCEPT
- WATER → surface / subsurface drainage
- GEOMETRY → controlled excavation and toe protection

INSTITUTIONAL OR COMMUNITY RESPONSE
- MATERIAL → retaining protection bioengineering
- LIMIT → site-specific design maintenance residual risk

CAPACITY / STATUS / EVIDENCE LIMIT
- WATER → surface / subsurface drainage
- GEOMETRY → controlled excavation and toe protection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MATERIAL → retaining protection bioengineering; LIMIT → site-specific design maintenance residual risk.

08

STAGE 08

ROUTE AND SETTLEMENT RESILIENCE

ROUTE AND SETTLEMENT RESILIENCE · SITING + SPOIL · DRAINAGE CONTROL · ALTERNATIVE ACCESS + SAFE STOPPAGE · ASSEMBLY AREA + EVACUATION ROUTE · HOSPITAL POWER TELECOM WATER CONTINUITY

ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL → ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE → ASSEMBLY AREA + EVACUATION ROUTE → HOSPITAL POWER TELECOM WATER CONTINUITY

HAZARD / RISK / RESILIENCE CONCEPT
- ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL
- ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

INSTITUTIONAL OR COMMUNITY RESPONSE
- ASSEMBLY AREA + EVACUATION ROUTE
- HOSPITAL POWER TELECOM WATER CONTINUITY

CAPACITY / STATUS / EVIDENCE LIMIT
- ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL
- ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL.

09

STAGE 09

WARNING-TO-ACTION CHAIN

WARNING-TO-ACTION CHAIN · COMMUNICATE UNCERTAINTY · CLOSE ROUTE · SAFE DESTINATION · VERIFY RECEIPT ACTION AND FEEDBACK · MUST REMEMBER · LANDSLIDE, AVALANCHE AND GLOF RISK IN THE HIMALAYA JOINS

MONITOR → ASSESS → FORECAST / BULLETIN → AUTHORISE → COMMUNICATE UNCERTAINTY → CLOSE ROUTE / EVACUATE → SAFE DESTINATION → VERIFY RECEIPT ACTION AND FEEDBACK → MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology

RISK CONDITION / INPUT
- MONITOR → ASSESS → FORECAST / BULLETIN
- AUTHORISE → COMMUNICATE UNCERTAINTY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- CLOSE ROUTE / EVACUATE → SAFE DESTINATION
- VERIFY RECEIPT ACTION AND FEEDBACK

LAST-MILE / INCLUSION / OUTCOME LIMIT
- MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology,...
- MONITOR → ASSESS → FORECAST / BULLETIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology,.

10

STAGE 10

DEVELOPMENT-RISK BALANCE

DEVELOPMENT-RISK BALANCE · CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS · SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK · RISK-INFORMED SITING NOT BLANKET CLAIMS · CLOSE DISTINCTION · SUSCEPTIBILITY IS NOT PROBABILITY; TRIGGER IS NOT ROOT VULNERABILITY

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS	VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS	CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS
SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK	CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability;...	SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

HAZARD / ACTOR / INSTRUMENT
- CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS
- SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

DECISIVE RISK OR MANDATE DISTINCTION
- VERDICT → RISK-INFORMED SITING NOT BLANKET CLAIMS
- CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability;...

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS
- SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability;.

11

STAGE 11

MOUNTAIN-RISK ANSWER SPINE

MOUNTAIN-RISK ANSWER SPINE · CLASSIFY PROCESS · SEPARATE CONDITION FROM TRIGGER · DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK · ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION · COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS · EVIDENCE LIMIT · SOURCE EVENT DATE, LOCATION, RAINFALL

CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER → DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK → ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION → COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall

HAZARD / RISK / RESILIENCE CONCEPT
- CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER
- DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

INSTITUTIONAL OR COMMUNITY RESPONSE
- ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION
- COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

CAPACITY / STATUS / EVIDENCE LIMIT
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall,...
- CLASSIFY PROCESS → SEPARATE CONDITION FROM TRIGGER

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · NDMA'S EXPLICIT NON-ISSUANCE OF CLOUDBURST-SPECIFIC GUIDELINES · STATE-LEVEL RESPONSIBILITY FOR AREA-SPECIFIC MEASURES · AVALANCHE WARNING SITS WITH A DEFENCE-RESEARCH INSTITUTION, NOT · MINING-LINKED LANDSLIDES AS A DISTINCT SUB-CATEGORY · THE UTTARAKHAND 2013 DISASTER AS A DOCUMENTED MULTI-CLOUDBURST · CLOUDBURST'S NEAR-IRREDUCIBLE PREDICTION GAP · VISIONIAS STATES

OPTIONAL RISK / RESILIENCE DEPTH
- NDMA's explicit non-issuance of cloudburst-specific guidelines
- State-level responsibility for area-specific measures ; landslide
- CAUTION: Avalanche warning sits with a defence-research institution, not
- Mining-linked landslides as a distinct sub-category . VisionIAS's

MANDATE, WARNING OR OUTCOME LIMIT
- The Uttarakhand 2013 disaster as a documented multi-cloudburst.
- CAUTION: Cloudburst's near-irreducible prediction gap . VisionIAS states
- ISRO's Himalayan glacial-lake monitoring role (PDF p. 79) is the
- CAUTION: Avalanche prediction shares cloudburst's short-lead-time

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- CAUTION: Himalayan/hill-state populations dependent on tourism and
- CAUTION: Hydropower/infrastructure development versus glacial-risk
- CAUTION: Terrace farming versus Jhumming (shifting cultivation) in the
- NOT: The "landslide vulnerability zones are mapped, therefore risk is

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels